

Benjamin Cohen-Stead

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PROFESSIONAL EXPERIENCE

2024–present Research Assistant Professor	University of Tennessee, Knoxville Department of Physics & Astronomy
2022–2024 Postdoctoral Research Associate Supervisor: Professor Steven Johnston	University of Tennessee, Knoxville Department of Physics & Astronomy
2020–2022 Graduate Student Researcher Supervisor: Dr. Kipton Barros	Los Alamos National Laboratory
2014–2016 Associate Data Scientist	Picarro, Inc.
Summer, 2014 Data Science Intern	Picarro, Inc.

EDUCATION

University of California, Davis Ph.D. in Physics Advisor: Professor Richard Scalettar	Davis, Ca 2016–2022
Whitman College B.A. in Physics	Walla Walla, Wa 2010–2014

GRANTS & AWARDS

Scientific Software Research Faculty Award Organization: Simons Foundation - Principle Investigator: Dr. Benjamin Cohen-Stead - Project: User-friendly and Extensible Quantum Monte Carlo Related Tools	2024–2029
UC-National Lab In-Residence Graduate Fellowship Organization: Los Alamos National Laboratory (LANL) - Supervisor: Dr. Kipton Barros - Project: Langevin Methods for Quantum Electron-Phonon Simulations	2020–2022

TALKS

Department of Physics Colloquium

April 7, 2026

Harvey Mudd College

Claremont, Ca

- Title: A Determinant Quantum Monte Carlo Study of a Minimal Electron-Phonon Model for Kekulé Lattice Distortions in Strained Graphene
- Host: Professor Eduardo Ibarra García Padilla

JuliaCon

July 25, 2025

Computational Chemistry and Materials Science Minisymposium

Pittsburgh, Pa

- Title: [Simulating Strongly-Correlated Material Models](#)

Department of Physics Seminar

November 8, 2023

San Jose State University

Department of Physics & Astronomy

- Title: Revisiting the bismuthate family of high- T_c superconductors with a modern tool kit
- Host: Professor Ehsan Khatami

Condensed Matter Physics Seminar

October 18, 2023

University of Tennessee, Knoxville

Department of Physics & Astronomy

- Title: Bond-Stretching Electron-Phonon Interactions in BaBiO_3
- Host: Professor Steven Johnston

OPEN SOURCE CODES

SmoQyElPhQMC.jl: <https://github.com/SmoQySuite/SmoQyElPhQMC.jl.git>

Package implementing a quantum Monte Carlo method for simulating electron-phonon models that scales near-linearly with system size.

SmoQyDQMC.jl: <https://github.com/SmoQySuite/SmoQyDQMC.jl.git>

Flexible implementation of the determinant quantum Monte Carlo method for simulating Hubbard and electron-phonon interactions.

SmoQyKPMCore.jl: <https://github.com/SmoQySuite/SmoQyKPMCore.jl.git>

A lightweight and flexible implementation of the Kernel Polynomial Method (KPM) for approximating functions of Hermitian matrices.

SmoQySynthAC.jl: <https://github.com/SmoQySuite/SmoQySynthAC.jl.git>

A package for generating synthetic noisy correlation function data intended to mimic that generated by a quantum Monte Carlo simulation. This package is useful for testing efficacy of various analytic continuation methods.

SmoQyHankelCorrCleaner.jl: <https://github.com/SmoQySuite/SmoQyHankelCorrCleaner.jl.git>

Julia package exporting methods for denoising imaginary time correlation data using the Hankel projection method introduced in [arXiv:2403.12349](#)

JDQMCFramework.jl: <https://github.com/SmoQySuite/JDQMCFramework.jl.git>

Julia package exporting a suite of types and routines useful for writing a determinant quantum Monte Carlo code.

JDQMCMeasurements.jl: <https://github.com/SmoQySuite/JDQMCMeasurements.jl.git>

Julia package implementing various correlation measurements that are frequently made in determinant quantum Monte Carlo simulations.

StableLinearAlgebra.jl: <https://github.com/SmoQySuite/StableLinearAlgebra.jl.git>

Exports numerically stable linear algebra routines used in determinant quantum Monte Carlo codes.

Checkerboard.jl: <https://github.com/SmoQySuite/Checkerboard.jl.git>

Implements the checkerboard approximation for representing exponentiated kinetic for tight-binding models.

LatticeUtilities.jl: <https://github.com/SmoQySuite/LatticeUtilities.jl.git>

Julia package for representing arbitrary periodic lattice geometries.

MuTuner.jl: <https://github.com/cohensbw/MuTuner.jl.git>

Implements a method for tuning the chemical potential in grand canonical Monte Carlo simulations to achieve a target particle density.

PATENTS

- [1] A. Nottrott, S. MacMullin, B. Cohen-Stead, C. W. Rella, and S. M. Tan, “Aggregate leak indicator display systems and methods”, Utility patent US10962437B1, 2021.

PREPRINTS

- [1] B. Cohen-Stead, S. M. Costa, J. Neuhaus, A. T. Ly, Y. Zhang, R. Scalettar, K. Barros, and S. Johnston, “SmoQyDQMC.jl: A flexible implementation of determinant quantum Monte Carlo for Hubbard and electron-phonon interactions (version 2.0 release)”, *arXiv preprint arXiv:2311.09395*, 2026. DOI: [10.48550/arXiv.2311.09395](https://doi.org/10.48550/arXiv.2311.09395).
- [2] C. Jordan, G. Issa, E. Khatami, R. Scalettar, B. Cohen-Stead, and S. Johnston, “Charge order in the half-filled bond-Holstein model”, *arXiv preprint arXiv:2601.13121*, 2026. DOI: [10.48550/arXiv.2601.13121](https://doi.org/10.48550/arXiv.2601.13121).

PUBLICATIONS

- [1] S. Malkaruge Costa, B. Cohen-Stead, and S. Johnston, “Antiferromagnetism and Kekulé valence bond order in the honeycomb optical Su-Schrieffer-Heeger-Hubbard model”, *Phys. Rev. B*, vol. 113, p. 115 138, 11 Mar. 2026. DOI: [10.1103/8p79-4r7x](https://doi.org/10.1103/8p79-4r7x).
- [2] M. Naamneh, E. C. O’Quinn, E. Paris, D. McNally, Y. Tseng, W. R. Pudłko, D. J. Gawryluk, J. Shamblin, B. Cohen-Stead, M. Shi, M. Radovic, M. K. Lang, T. Schmitt, S. Johnston, and N. C. Plumb, “Persistence of small polarons into the superconducting doping range of $\text{Ba}_{1-x}\text{K}_x\text{BiO}_3$ ”, *Phys. Rev. Res.*, vol. 7, p. 043 082, 4 Oct. 2025. DOI: [10.1103/s3p1-cy1s](https://doi.org/10.1103/s3p1-cy1s).
- [3] Y. Zhang, P. M. Dee, B. Cohen-Stead, T. A. Maier, S. Johnston, and R. Scalettar, “Optimizing the critical temperature and superfluid density of a metal-superconductor bilayer”, *Phys. Rev. B*, vol. 112, p. 064 510, 6 Aug. 2025. DOI: [10.1103/lcgr-bqcv](https://doi.org/10.1103/lcgr-bqcv).
- [4] A. Tanjaroon Ly, B. Cohen-Stead, and S. Johnston, “Antiferromagnetic and bond-order-wave phases in the half-filled two-dimensional optical Su-Schrieffer-Heeger-Hubbard model”, *Phys. Rev. B*, vol. 111, p. 245 138, 24 Jun. 2025. DOI: [10.1103/2bnf-tmtc](https://doi.org/10.1103/2bnf-tmtc).
- [5] P. Mai, B. Cohen-Stead, T. A. Maier, and S. Johnston, “Fluctuating charge-density-wave correlations in the three-band Hubbard model”, *Proceedings of the National Academy of Sciences*, vol. 121, no. 50, e2408717121, 2024. DOI: [10.1073/pnas.2408717121](https://doi.org/10.1073/pnas.2408717121). eprint: <https://www.pnas.org/doi/pdf/10.1073/pnas.2408717121>.

- [6] J. Neuhaus, N. S. Nichols, D. Banerjee, B. Cohen-Stead, T. A. Maier, A. D. Maestro, and S. Johnston, “SmoQyDEAC.jl: A differential evolution package for the analytic continuation of imaginary time correlation functions”, *SciPost Phys. Codebases*, p. 39, 2024. DOI: [10.21468/SciPostPhysCodeb.39](https://doi.org/10.21468/SciPostPhysCodeb.39).
- [7] S. Malkaruge Costa, B. Cohen-Stead, and S. Johnston, “Kekulé valence bond order in the honeycomb lattice optical Su-Schrieffer-Heeger model and its relevance to graphene”, *Phys. Rev. B*, vol. 110, p. 115 130, 11 Sep. 2024. DOI: [10.1103/PhysRevB.110.115130](https://doi.org/10.1103/PhysRevB.110.115130).
- [8] B. Cohen-Stead, S. M. Costa, J. Neuhaus, A. T. Ly, Y. Zhang, R. Scalettar, K. Barros, and S. Johnston, “SmoQyDQMC.jl: A flexible implementation of determinant quantum Monte Carlo for Hubbard and electron-phonon interactions”, *SciPost Phys. Codebases*, p. 29, 2024. DOI: [10.21468/SciPostPhysCodeb.29](https://doi.org/10.21468/SciPostPhysCodeb.29).
- [9] A. Tanjaroon Ly, B. Cohen-Stead, S. Malkaruge Costa, and S. Johnston, “Comparative study of the superconductivity in the holstein and optical su-schrieffer-heeger models”, *Phys. Rev. B*, vol. 108, p. 184 501, 18 Nov. 2023. DOI: [10.1103/PhysRevB.108.184501](https://doi.org/10.1103/PhysRevB.108.184501).
- [10] S. Malkaruge Costa, B. Cohen-Stead, A. T. Ly, J. Neuhaus, and S. Johnston, “Comparative determinant quantum Monte Carlo study of the acoustic and optical variants of the Su-Schrieffer-Heeger model”, *Phys. Rev. B*, vol. 108, p. 165 138, 16 Oct. 2023. DOI: [10.1103/PhysRevB.108.165138](https://doi.org/10.1103/PhysRevB.108.165138).
- [11] O. Bradley, B. Cohen-Stead, S. Johnston, K. Barros, and R. T. Scalettar, “Charge order in the kagome lattice Holstein model: A hybrid Monte Carlo study”, *npj Quantum Materials*, vol. 8, no. 1, p. 21, 2023. DOI: [10.1038/s41535-023-00553-y](https://doi.org/10.1038/s41535-023-00553-y).
- [12] S. Karakuzu, B. Cohen-Stead, C. D. Batista, S. Johnston, and K. Barros, “Flexible class of exact hubbard-stratonovich transformations”, *Phys. Rev. E*, vol. 107, p. 055 301, 5 May 2023. DOI: [10.1103/PhysRevE.107.055301](https://doi.org/10.1103/PhysRevE.107.055301).
- [13] B. Cohen-Stead, K. Barros, R. Scalettar, and S. Johnston, “A hybrid Monte Carlo study of bond-stretching electron-phonon interactions and charge order in BaBiO₃”, *npj Computational Materials*, vol. 9, no. 1, p. 40, 2023. DOI: [10.1038/s41524-023-00998-6](https://doi.org/10.1038/s41524-023-00998-6).
- [14] P. M. Dee, B. Cohen-Stead, S. Johnston, and P. J. Hirschfeld, “Charge correlations suppress unconventional pairing in the Holstein model”, *Phys. Rev. B*, vol. 107, p. 104 503, 10 Mar. 2023. DOI: [10.1103/PhysRevB.107.104503](https://doi.org/10.1103/PhysRevB.107.104503).
- [15] B. Cohen-Stead, O. Bradley, C. Miles, G. Batrouni, R. Scalettar, and K. Barros, “Fast and scalable quantum Monte Carlo simulations of electron-phonon models”, *Phys. Rev. E*, vol. 105, p. 065 302, 6 Jun. 2022. DOI: [10.1103/PhysRevE.105.065302](https://doi.org/10.1103/PhysRevE.105.065302).
- [16] C. Miles, B. Cohen-Stead, O. Bradley, S. Johnston, R. Scalettar, and K. Barros, “Dynamical tuning of the chemical potential to achieve a target particle number in grand canonical Monte Carlo simulations”, *Phys. Rev. E*, vol. 105, p. 045 311, 4 Apr. 2022. DOI: [10.1103/PhysRevE.105.045311](https://doi.org/10.1103/PhysRevE.105.045311).
- [17] G. Paleari, F. Hébert, B. Cohen-Stead, K. Barros, R. Scalettar, and G. G. Batrouni, “Quantum Monte Carlo study of an anharmonic Holstein model”, *Phys. Rev. B*, vol. 103, p. 195 117, 19 May 2021. DOI: [10.1103/PhysRevB.103.195117](https://doi.org/10.1103/PhysRevB.103.195117).
- [18] B. Cohen-Stead, K. Barros, Z. Meng, C. Chen, R. T. Scalettar, and G. G. Batrouni, “Langevin simulations of the half-filled cubic Holstein model”, *Phys. Rev. B*, vol. 102, p. 161 108, 16 Oct. 2020. DOI: [10.1103/PhysRevB.102.161108](https://doi.org/10.1103/PhysRevB.102.161108).
- [19] B. Cohen-Stead, N. C. Costa, E. Khatami, and R. T. Scalettar, “Effect of strain on charge density wave order in the Holstein model”, *Phys. Rev. B*, vol. 100, p. 045 125, 4 Jul. 2019. DOI: [10.1103/PhysRevB.100.045125](https://doi.org/10.1103/PhysRevB.100.045125).